

# MADLINE C. BUSSE

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## EDUCATION

- The University of Chicago** | Chicago, IL 2021–2025  
*Bachelor of Arts, Double Major in Computer Science and Philosophy & Allied Fields.*
- **GPA:** 3.676/4.000 | **National Merit Scholar** | **SAT:** 1590 (800M, 790R).
  - **Relevant Coursework:** Algorithms, Data Science, Discrete Math, HCI, Security & Privacy, Systems, Robotics.
- MIT Sloan School of Management** | Cambridge, MA Eligible 2027–2030  
*Deferred MBA Admit, Early Admission Program.*
- AI Ethics Seminar at Duke University** | Durham, NC June 2024
- Sole undergraduate selected for intensive seminar on AI design ethics.

## EXPERIENCE

- PMG, AI Software Engineer** | Dallas, TX June 2025–present
- Partnered with **Nike North America** integrated media team to **build Python-based automation** for Meta/Snapchat ad APIs, saving 3+ hours/week.
  - Created **end-to-end testing suites in Cypress** for micro-frontends, collaborating with UX/Product to identify and test expected user flows.
  - Designed and implemented full-stack **data pipelines** for TikTok and Pinterest trend APIs, surfacing in **real-time Looker dashboards** for non-technical stakeholders to visualize trending searches.
  - Automated daily reporting systems by building data pipelines integrated with **AI processing** and scheduled delivery to **slack channels**.
- Zenblen, Front-End Software Engineer Intern** | Chicago, IL July 2024–September 2024
- Led an **end-to-end UI redesign** of a **robotic kiosk**, prototyping iteratively in **Figma** and implementing features via **Dart** to increase interactivity and accessibility of the kiosk.
  - Partnered with CEO and marketing to align system UI with **growth initiatives**, incorporating a loyalty program and brand identity into the kiosk through interactive pop-ups.
- Human Robot Interaction Lab, Research Intern** | Chicago, IL June 2023–August 2023
- Integrated **NAO robot** with **ChatGPT API** and **ROS control loops** to enable real-time speech and motion.
  - Engineered automated statistical pipelines in **Python and R** for human-subject experiments (150+ participants).

## TECHNICAL PROJECTS ([github.com/madbusse](https://github.com/madbusse))

- Interactive CTA Map** January 2025–February 2025
- Programmed a **swarm of Toio robots** to simulate an interactive CTA map, translating the physical testing environment into a digital simulation and 3D-printing system elements using **Fusion360**.
- DanceBot** April 2024–May 2024
- Integrated a real-time ML system into a **Turtlebot3** to recognize sign language with **98% accuracy** using **OpenCV** and **TensorFlow** and perform corresponding dance moves via **ROS control loops**.

## PUBLICATIONS

**The Impact of a Robot's Agreement (or Disagreement) on Human-Human Interpersonal Closeness in a Two-Person Decision-Making Task.** *Ting-Han Lin, Yuval Rubin Kopelman, Madeline Busse, Sarah Sebo, Hadas Erel.* Computers in Human Behavior. *Accepted, forthcoming 2026.*

## COMMUNITY INVOLVEMENT

- Dallas Animal Services, Volunteer** August 2025–present
- Enrich animal welfare by organizing activities outside the shelter for 4+ hours monthly.
- By the Hand Club for Kids, Volunteer** November 2023–October 2024
- Fundraised over **\$1,000** for the 2024 Chicago Marathon Team while leading monthly afterschool programs for kids.

## SKILLS & CERTIFICATIONS

- **Languages:** Python, C, R, Dart, JavaScript, SQL, HTML/CSS.
- **Frameworks & Libraries:** React.js, Pandas, Numpy, Flutter, TensorFlow, PyTorch, Scikit-learn, Tkinter.
- **Cloud & Development:** AWS (S3, Lambda), Git, Figma, Docker, Looker Studio, Jira, Confluence, Linux.
- **Robotics & VR:** ROS, OpenCV, Gazebo, Unity, Oculus, Arduino IDE, Fusion360.